



## Understanding Solid Electrolyte Interface Film Formation on Graphite Electrodes

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Using ac impedance, we studied the formation process of solid electrolyte interface (SEI) film on graphite electrode during initial cycles. Results show that the SEI formation takes place through two major stages. The first stage takes place at voltages above 0.25 V (before lithiation of graphite), during which a loose and highly resistive film is formed. The second stage occurs at a narrow voltage range of 0.25-0.04 V, which proceeds simultaneously with lithiation of graphite electrode. In the second stage, a stable, compact, and highly conductive SEI film is produced.

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Formation is a process to form a protective solid electrolyte interface (SEI) film on the surface of graphite electrode, which is usually completed during the first few cycles. The formed SEI film plays a major role in determining battery performance, including cycle life, calendar life (self-discharge), safety, faradaic efficiency, and irreversible capacity.<sup>1,2</sup> Poor cycling behavior of lithium-ion batteries at both elevated temperature ( $> 50^{\circ}\text{C}$ ) and low temperature ( $< -30^{\circ}\text{C}$ ) are closely associated with the instability (at high temperature)<sup>3</sup> and low ionic conductivity (at low temperature)<sup>2,4</sup> of the SEI films. Therefore, in the past few years many researches have focused on identifying and understanding these SEI films. For this purpose, some models have been proposed to explain the mechanism of SEI formation.<sup>5-8</sup> All these models involve a reductive decomposition of electrolyte components (solvents and salt), which generally takes place during the first few cycles. SEI films on the surfaces of graphite electrode consist of many insoluble materials including  $\text{Li}_2\text{O}$ ,  $\text{LiF}$ ,  $\text{Li}_2\text{CO}_3$ ,  $\text{RCO}_2\text{Li}$ , alkoxides, and nonconducting polymers. In these materials the anions originate from the reductive decomposition of electrolyte components and the lithium ions arise from electrolyte or electrode. The formation process will permanently consume lithium ions and result in an irreversible capacity. The type of carbon, salt, and solvents significantly affects irreversible capacity of the graphite electrode and the reductive decomposition of solvents mostly occurs in a potential range of 0.4-0.9 V vs.  $\text{Li}^+/\text{Li}$ . Therefore, work was previously conducted aimed to improve the stability of SEI film and to reduce irreversible capacity, which occurs at 0.4-0.9 V, by introducing electrolyte additive or by modifying surface of natural graphites.<sup>9-11</sup> However, little detailed work has been published concerning the potential range in which SEI is formed and how the cell potential affects the stability of SEI films.

In this work, we used electrochemical impedance spectroscopy (EIS) to measure the resistance of SEI film during initial forming cycles and analyzed the correlation between resistance of SEI film and its formation potential. The formation process of SEI film on graphite electrode is discussed.

### Experimental

Solvents ethylene carbonate (EC) (99.95%) and propylene carbonate (PC) (99.98%) from Grant Chemical, ethylene methyl carbonate (EMC) (99.9%) from EM Industries, Inc., and  $\text{LiPF}_6$  from Stella Chemifa Corp. were used as received. Two electrolytes (1 M  $\text{LiPF}_6$  1:1:3 PC/EC/EMC and 1 M  $\text{LiPF}_6$  3:7 EC/EMC, weight ratio) were prepared in an argon-filled glove box. Water content of the obtained electrolytes was determined by Karl-Fisher titration to be 10-15 ppm. SAFT America supplied us with graphite

electrode film with a designed specific capacity of  $0.93 \text{ mAh/cm}^2$  under the cycling conditions of 2.5-3.9 V in a Li-ion cell. While our measured capacity in a Li half-cell was  $1.41 \text{ mAh/cm}^2$  at 0.5 mA/cm<sup>2</sup> between 0.002 and 1.0 V. Li/graphite button cells with an electrode area of  $1.27 \text{ cm}^2$  were assembled in the same glovebox and stored for 24 h before test.

Cells were cycled on a Maccor series 4000 tester at 0.5 mA/cm<sup>2</sup> between 1.0 and 0.02 V except for the first cycle and tenth cycle, at which the current rate was 0.05 mA/cm<sup>2</sup>. AC impedance measurement and cell cycling of the first and tenth cycle were performed with a Solartron SI 1287 electrochemical interface and an SI 1260 impedance/gain-phase analyzer using CorrWare and Zplot software, and the obtained data were analyzed with CorrView and ZView softwares. Cell impedance was potentiostatically measured by applying an ac bias of 5 mV amplitude over the frequency of 100 kHz to 0.01 Hz after the cell had been cycled to a potential at 0.05 mA/cm<sup>2</sup> and attained at this potential for 20 min.

### Results and Discussion

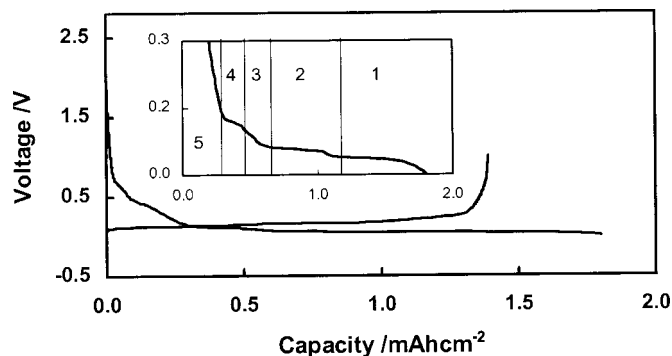
The impedance of a Li/graphite cell consists of the impedances of electrolyte, electrodes, and passive films on the electrodes. To minimize the effect of the lithium counter electrode on the cell impedance, it is necessary to stabilize the passive film grown on the surface of the lithium electrode. We measured impedance of a newly assembled Li/graphite cell at different storage time at room temperature, and found that the cell's EIS no longer changed after storing for 24 h. Therefore, we stored all cells for 24 h before measurements were made. Figure 1 shows voltage profile of a Li/graphite cell vs. capacity of the graphite electrode for the first lithiation-delithiation cycle at a current density of  $0.05 \text{ mA/cm}^2$ . Electrolyte used in the cell was a solution of 1 M  $\text{LiPF}_6$  1:1:3 PC/EC/EMC (wt). The inset in Fig. 1 is part of the voltage-capacity curve during the first lithiation process, from which one may find five stages for the lithiation of graphite electrode in the voltage range below 0.3 V. According to the previous study,<sup>12</sup> these stages are ascribed to five continuous transitions:  $\text{LiC}_{72} \rightleftharpoons \text{LiC}_{36} \rightleftharpoons \text{LiC}_{27} \rightleftharpoons \text{LiC}_{18} \rightleftharpoons \text{LiC}_{12} \rightleftharpoons \text{LiC}_6$ . It was calculated from Fig. 1 that the coulombic efficiency of delithiation during the first cycle is 77% and irreversible capacity is 23%. According to the literature,<sup>5-8</sup> this irreversible capacity is contributed to the formation of SEI film on the graphite. Among 23% of the irreversible capacity, 11% took place at the voltages of above 0.25 V (before lithium intercalation) and the other 12% at the voltages of between 0.25 and 0.04 V. In other words, 12% of the irreversible capacity is attributed to the SEI film developed during the intercalation of lithium ions into graphite.

A general understanding is that irreversible capacity of lithium ion intercalation into the graphite electrode during the first few cycles is associated with the reductive decomposition of solvents and the simultaneous formation of SEI film. To observe in what

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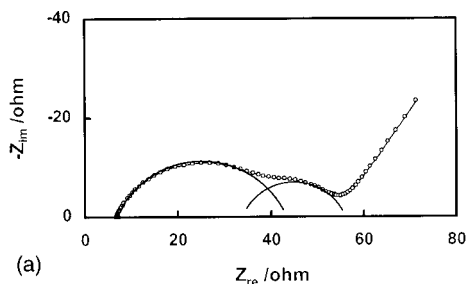
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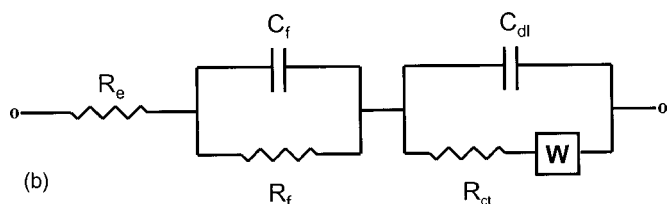
**Figure 1.** Voltage-capacity curves of Li/graphite cell for the first cycle at 0.05 mA/cm<sup>2</sup>. Electrolyte was a 1 M LiPF<sub>6</sub> 1:1:3 PC/EC/EMC solution (wt).

voltage ranges a SEI film is formed, we studied the correlation of cell impedance and cell voltages by measuring impedance at various voltages. Note that the voltage corresponds well to the composition ( $x$ ) of lithium graphite compound ( $\text{Li}_x\text{C}_6$ ) as it is below 0.25 V. Figure 2a shows typical impedance responses recorded after the graphite electrode was first lithiated to 0.02 V and then delithiated back to 0.25 V during the first cycle in a 1 M LiPF<sub>6</sub> 1:1:3 PC/EC/EMC electrolyte. EIS of the Li/graphite cell consists of two overlapped semicircles at high and medium frequency range, and a spike at low frequency range. Figure 2b is an equivalent circuit used to analyze the electrochemical process for the lithium intercalation into graphite electrode.<sup>13-16</sup>  $R_e$  represents the total resistances of electrolyte, electrode, and separator.  $R_f$  and  $C_f$  are the resistance and capacitance of the SEI film, respectively, and  $R_{ct}$  and  $C_{dl}$  are the charge-transfer resistance and double-layer capacitance, respectively.  $W$  is the Warburg impedance arising from the semi-infinite diffusion of the lithium ions in graphite electrode.

Figure 3 shows dependence of  $R_e$ ,  $R_f$ , and  $R_{ct}$  on cell voltages for the first lithiation-delithiation cycle of the Li/graphite cell using 1 M LiPF<sub>6</sub> 1:1:3 PC/EC/EMC electrolyte. As shown in Fig. 3a, the resistance ( $R_e$ ) of electrode and electrodes is nearly independent of the process of lithiation and delithiation. However, the SEI resistance ( $R_f$ ) presents an interesting change with the voltage (see Fig.

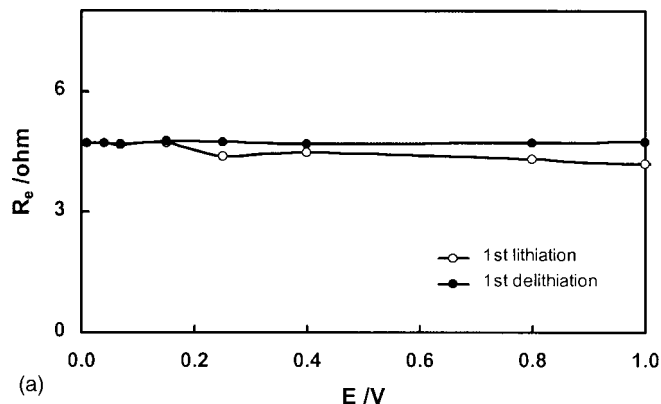


(a)

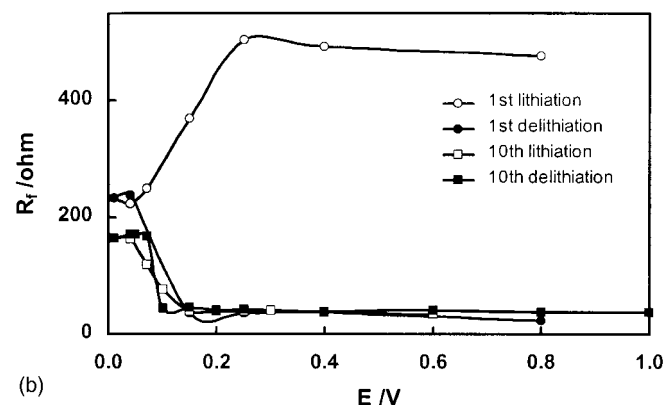


(b)

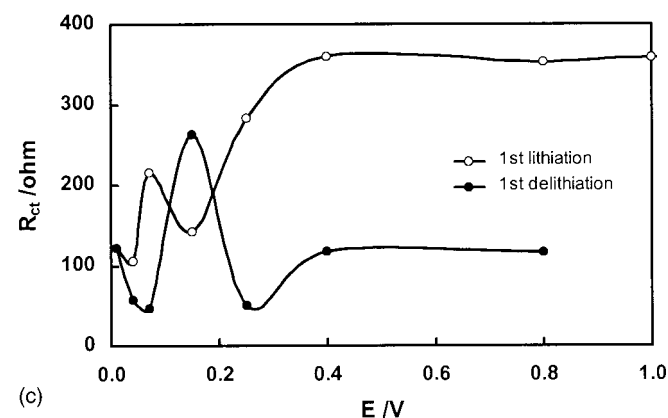
**Figure 2.** (a) Complex impedance plot of Li/graphite cell at 0.25 V obtained from the first cycle after graphite electrode was lithiated to 0.02 V and delithiated back to 0.25 V at 0.05 mA/cm<sup>2</sup>. (b) Equivalent circuit used for the analysis of the complex impedance plots.



(a)



(b)



(c)

**Figure 3.** Resistance as a function of cell voltage for the first cycle of Li/graphite cell. (a) Resistance ( $R_e$ ) of electrolyte and electrode, (b) resistance ( $R_f$ ) of the SEI film, and (c) resistance ( $R_{ct}$ ) of charge transfer. Note that  $R_f$  of the tenth cycle is plotted in (b).

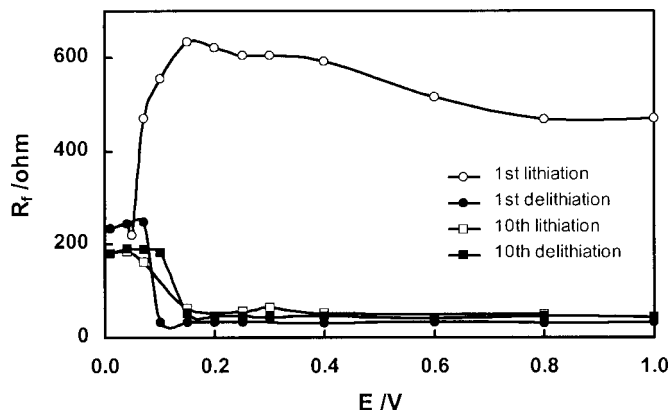
3b). In the voltages of above 0.25 V,  $R_f$  stayed at a high level (450-510  $\Omega$ ) and slowly increased as the voltage falls. Cycling data shows that this process produced 11% of irreversible capacity, which is believed to be associated with the solvent decomposition. We think that the high value and only a slight increase of the  $R_f$  above 0.25 V is probably because most of the products of solvent reductive decomposition are soluble in the electrolyte and the insoluble products compose a loose and resistive SEI film. The SEI film formed at this stage may be not strong enough to protect graphite electrode. More interestingly,  $R_f$  value sharply decreases from 500 to 220  $\Omega$  in a narrow voltage range of 0.25 and 0.04 V, at which lithiation of graphite electrode is undergoing. From the fact that the lithiation process produces 12% of irreversible capacity and  $R_f$  value sharply decreases, one may conclude that this process (0.25-0.04 V) devel-

ops a compact and conductive SEI film. It is probable that only the SEI film formed between 0.25 and 0.04 V is indeed protective to both graphite electrode and electrolyte solvents.

It is surprising that the  $R_f$  value further decreases and the drop in it is so large while the graphite electrode is undergoing delithiation in the voltage range of 0.04–0.25 V (Fig. 3b). This phenomenon is likely associated with one or two factors; SEI film physical restructure caused by volume change of the graphite electrode during lithium deintercalation and SEI film dissolution into the electrolyte. The former will result in a reversible voltage dependence of the  $R_f$  value, while the latter will produce a permanent decrease in the  $R_f$ . To clarify which mechanism dominates the decrease of  $R_f$  during delithiation, we measured  $R_f$  value of the cell of tenth cycle, at which the formation of SEI film is expected to have been completed. Note that the delithiation capacity of graphite electrode lost 4% after 160 cycles, but had no change between the first cycle and the tenth cycle. The  $R_f$  value of tenth cycle is also plotted in Fig. 3b, which shows that the  $R_f$  value changes reversibly between lithiation and delithiation. Therefore, it could be concluded that the reversible sharp drop in the  $R_f$  value during delithiation from 0.04 to 0.25 V originates from physical restructure of the SEI film. A permanent loss of 69  $\Omega$  in the  $R_f$  value between 0.04 and 0 V is also observed from the first to tenth cycles (Fig. 3b). This permanent loss in the  $R_f$  value is probably due to the SEI film becoming more conductive as it trends complete with the increase of cycle number during the initial forming cycles.

Figure 3c plots  $R_{ct}$  value as a function of the voltage. In general,  $R_{ct}$  is more dependent on the composition ( $x$ ) of graphite electrode ( $\text{Li}_x\text{C}_6$ ). As shown in Fig. 3c,  $R_{ct}$  value has less change in the voltage range above 0.4 V (before lithiation). Slightly dramatic change of  $R_{ct}$  values is observed from the voltages of below 0.4 V. In both lithiation and delithiation processes,  $R_{ct}$  changes vs. the cell voltage exhibit two minimum peaks with a little shift in the peak voltages. This phenomenon is related to the multistage processes of lithiation and delithiation, in which some stages cannot be reflected from the change of  $R_{ct}$  value. Similar result was previously reported by Kim and Park but without further explanation.<sup>17</sup> However, it has been reported that the diffusivity of lithium ions within graphite varies from stage to stage of the lithium ion intercalation, and a lower diffusivity often appears at the regions of stage transition.<sup>18–20</sup> It is noticed from Fig. 3c that  $R_{ct}$  value of the lithiation is larger than that of the delithiation. This is related to the fact that the lithiation has less diffusivity of lithium ions within graphite electrode, as compared to the delithiation.<sup>20</sup>

To see whether the results obtained above are universal in the graphite electrodes, we measured impedance of the Li/graphite cell using a 1 M  $\text{LiPF}_6$  3:7 EC/EMC electrolyte at various cycling stages. Testing conditions and cell configurations are the same as used in the above experimental. Coulombic efficiency of the first delithiation in 1 M  $\text{LiPF}_6$  3:7 EC/EMC electrolyte was 87%. It was evaluated that 7% of irreversible capacity was produced before lithiation (above 0.25 V) and 6% was raised from the voltages of below 0.25 V (during lithiation). Irreversible capacity above 0.25 V was likely formed on the edges of the graphite particles, and varies with surface conditions of the graphite. Therefore, this irreversible capacity is highly dependent on the type of carbons, salts, solvents, and even a small amount of additives.<sup>1,9–11</sup> Dependence of the  $R_e$ ,  $R_f$ , and  $R_{ct}$  on the cell voltage (or, composition  $x$  in  $\text{Li}_x\text{C}_6$  that corresponds to the cell voltage well) in 1 M  $\text{LiPF}_6$  3:7 EC/EMC electrolyte is very similar to that observed from 1 M  $\text{LiPF}_6$  1:1:3 PC/EC/EMC. In particular, Fig. 4 plots  $R_f$  of the Li/graphite cell using 1 M  $\text{LiPF}_6$  3:7 EC/EMC electrolyte as a function of the voltage for the first and tenth lithiation-delithiation cycles. Comparing Fig. 3b and 4, one finds that two-stage formation



**Figure 4.** Response of the resistance ( $R_f$ ) of the SEI film to the cell voltage for the Li/graphite cell using 1 M  $\text{LiPF}_6$  3:7 EC/EMC (wt) electrolyte during the first and tenth cycles at 0.05  $\text{mA}/\text{cm}^2$ .

of the SEI film is popular in the graphite electrodes: one is before lithiation and the other one is during lithiation. Both stages involve a reductive decomposition of the electrolytes solvents.

### Conclusion

AC impedance study shows that SEI film is formed on the graphite electrode through two major stages. The first stage undergoes at voltages of above 0.25 V, and the second stage between 0.25 and 0.04 V during the intercalation of lithium ions into graphite electrode. Protective and conductive SEI film is formed in the second stage.

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